EXERCISE 5

Integrating Factor for a Non-Exact Equation

Original statement · ProblemsCh2.pdf, p. 2

Non-exact • Integrating factor μ(x)

PROBLEM STATEMENT

> Find an integrating factor depending only on x that makes the equation
>
> $$3xy + y^2 + (x^2 + xy)\frac{dy}{dx} = 0$$
>
> exact, and find its solution.

Source: ProblemsCh2.pdf, Exercise 5 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Write the equation as $M + N\,y' = 0$ with

$$M(x,y) = 3xy + y^2, \qquad N(x,y) = x^2 + xy = x(x+y).$$

M, N and their partial derivatives are polynomials, hence continuous on $\mathbb{R}^2$. Exactness test:

$$M_y = 3x + 2y, \qquad N_x = 2x + y.$$

$M_y - N_x = x + y$, which is not identically zero, so the equation is not exact. We look for $\mu = \mu(x)$ (nowhere zero on the domain considered) such that $\mu M + \mu N\,y' = 0$ is exact. We work on a region where $x \neq 0$ (the line $x = 0$ is excluded because $N(0,y) = 0$ there).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: multiplying the equation by a nonzero function does not change its solution set (on the region where the factor is nonzero) but can restore exactness. The requirement is $(\mu M)_y = (\mu N)_x$. If μ depends only on x, then $(\mu M)_y = \mu M_y$ and $(\mu N)_x = \mu' N + \mu N_x$, so the condition becomes

$$\mu M_y = \mu' N + \mu N_x \quad \Longleftrightarrow \quad \frac{\mu'}{\mu} = \frac{M_y - N_x}{N}.$$

This ordinary differential equation for $\mu(x)$ is solvable only if the right-hand side depends on x alone (otherwise no integrating factor of the form $\mu(x)$ exists). Then we integrate it, and continue with the standard exact-equation construction $F_x = \mu M$, $F_y = \mu N$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 4]

The class notes solve exact equations only; the search for an integrating factor $\mu(x)$ that makes a non-exact equation exact is a complement (not in the class notes). Once μ is found, $\mu M + \mu N\, y' = 0$ is solved with the construction of the notes: test $f_y = g_x$ for $f = \mu M$, $g = \mu N$, integrate f in x and add $\varphi(y)$, and fix φ from $F_y = g$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: find $\mu(x)$

Compute the quotient, factoring $N = x(x+y)$:

$$\frac{M_y - N_x}{N} = \frac{x+y}{x(x+y)} = \frac{1}{x} \qquad (x + y \neq 0).$$

This depends only on x, so μ exists. Solve $\mu'/\mu = 1/x$:

$$\int \frac{d\mu}{\mu} = \int \frac{dx}{x} \implies \ln|\mu| = \ln|x| + c \implies \mu(x) = x \quad \text{(taking the constant multiple equal to 1)}.$$

Step 2: check exactness of the new equation

Multiply by $\mu = x$:

$$\tilde{M} = xM = 3x^2y + xy^2, \qquad \tilde{N} = xN = x^3 + x^2y.$$

$$\tilde{M}_y = 3x^2 + 2xy, \qquad \tilde{N}_x = 3x^2 + 2xy.$$

They coincide, so the new equation is exact.

Step 3: potential function

Integrate $F_x = \tilde{M}$ in x at fixed y:

$$F = \int (3x^2y + xy^2)\,dx + h(y) = x^3y + \frac{x^2y^2}{2} + h(y).$$

Impose $F_y = \tilde{N}$:

$$F_y = x^3 + x^2y + h'(y) = x^3 + x^2y \implies h'(y) = 0.$$

So h is constant and the solution in implicit form is

$$x^3y + \frac{1}{2}x^2y^2 = C \qquad \Longleftrightarrow \qquad x^2y^2 + 2x^3y = K, \quad K = 2C.$$

Step 4: explicit form

Complete the square in y: $x^2y^2 + 2x^3y = x^2\left[(y+x)^2 - x^2\right]$. Then $x^2(y+x)^2 = K + x^4$, so

$$y(x) = -x \pm \frac{\sqrt{x^4 + K}}{|x|}, \qquad x \neq 0,\ x^4 + K \geq 0.$$

Remark: on the line $y = -x$ the coefficient $N = x(x+y)$ vanishes. Direct substitution of $y = -x$ gives $M = -3x^2 + x^2 = -2x^2$ and $N = 0$, so $M + Ny' = -2x^2 \neq 0$ and $y = -x$ is not a solution.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the potential: $F = x^3y + \frac{1}{2}x^2y^2$ has $F_x = 3x^2y + xy^2 = x(3xy + y^2) = xM$ and $F_y = x^3 + x^2y = x(x^2 + xy) = xN$, as required.

Check along a solution curve: implicit differentiation of $x^3y + \frac{1}{2}x^2y^2 = C$ gives $(3x^2y + xy^2) + (x^3 + x^2y)y' = 0$. Dividing by $x \neq 0$ gives back $(3xy + y^2) + (x^2 + xy)y' = 0$, the original equation. The division by x is why the line $x = 0$ must be excluded from the region of validity.

Numeric spot check of the explicit branch: with $K = 3$ and $x = 1$, $y = -1 + \sqrt{1 + 3} = 1$; then $F = 1 + \frac{1}{2} = \frac{3}{2} = K/2$, consistent. The slope from the ODE is $y' = -\frac{3xy + y^2}{x^2 + xy} = -\frac{4}{2} = -2$, and the derivative of $y = -x + \sqrt{x^4 + 3}/x$ is $-1 + \frac{x^4 - 3}{x^2\sqrt{x^4 + 3}}$, which equals $-1 + \frac{-2}{1 \cdot 2} = -2$ at $x = 1$.

RESULT

Integrating factor $\mu(x) = x$ (up to a constant factor). Solution: $x^3y + \frac{1}{2}x^2y^2 = C$ on regions with $x \neq 0$.